Satpreet Makhija

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Summary

Software engineer with experience building cloud infrastructure and distributed systems at Google Cloud and Firebolt. Background in Kubernetes, platform engineering, and reliable systems, complemented by research in program synthesis and formal verification.

Experience

Ashoka University — Sonipat, Haryana
Research Associate · Programming Languages — *Jun 2026 - Present*

- Designed and evaluated constraint-solving algorithms for program synthesis, using SAT-based optimization to generate programs satisfying hard constraints while optimizing soft objectives; co-authored a research manuscript under review.
- Developed a formal semantics for causal interventions and mechanized correctness results in Lean 4, including correspondence between graph transformations and structural causal models.

Firebolt — Bangalore, India
Software Engineer · Self-hosted Platform — *Nov 2025 - Mar 2026*

- Developed infrastructure components for Firebolt's self-hosted enterprise platform, supporting deployments in air-gapped environments.
- Implemented feature flag management and integrated Temporal-based workflow orchestration for platform operations.

Google — Bangalore, India
Software Engineer · Google Cloud, Distributed Cloud Team — *Mar 2024 - Sep 2025*

- Built infrastructure for Google Distributed Cloud, enabling Google Cloud services to run in air-gapped, sovereign, and regulated environments without public-network dependencies.
- Designed a Harbor-based artifact management system for a portable edge data center, developing Kubernetes controllers and custom CRDs for node-local image mirrors, self-healing, signature verification, and offline bootstrap.
- Developed Kubernetes controllers for observability CRDs that automatically provisioned Prometheus monitoring, alerting rules, and Grafana dashboards.
- Mentored junior engineers and supervised a summer intern who transitioned to a full-time role.

Google — Bangalore, India
Strategic Cloud Engineer · Google Cloud — *Jul 2022 - Mar 2024*

- Designed and implemented cloud infrastructure solutions for enterprise customers, providing hands-on engineering and architecture guidance for Google Cloud adoption.
- Prototyped Data Insights, an LLM-powered natural-language interface for tabular data discovery; the capability was subsequently incorporated into a Google Cloud data management product.
- Developed reusable infrastructure and data engineering tooling, including Terraform modules for Kubernetes and VPC deployments and open-source Dataproc templates for Spark/Hadoop ETL workloads.

Technical Skills

Languages: Go, Python, TypeScript, JavaScript, SQL, Nix, Lean 4, Rocq
Infrastructure & Systems: Kubernetes, Docker, Helm, Linux, Terraform, Google Cloud, distributed systems, gRPC, REST, CI/CD
Data & Observability: PostgreSQL, Prometheus, Grafana, OpenTelemetry, BigQuery, Spark, Hadoop
Research & Verification: Program synthesis, SAT-based algorithms, formal verification, programming-language semantics

Selected Research

Graph Surgery and the Do-Operator: A Precise Correspondence for Acyclic Structural Causal Models. Satpreet Makhija. *arXiv:2608.17634 [cs.AI]*, Aug 2026. arXiv · PDF · Lean 4 formalization

Education

Ashoka University — Sonipat, Haryana
Bachelor of Science (Honours) in Computer Science; Major CGPA: 3.74/4 — *Aug 2018 - Apr 2022*